For example, a 13x13 pixel map in a 3x3 convolution operation from 192 input maps to 192 output maps (let's say, in Alexanet layer 3) would require approximately 4MB of weight data and approximately 0.1 MB of input data from memory. As for bandwidth, this operation might consume about 3.2 GB/s on a 128 G-ops/s system. This is because the same input data is used to compute 192 outputs.

On the other hand, if you were to use an RNN (Recurrent Neural Network) then your bandwidth needs would be excessive. For example, systems such as Deep Speech 2 use 4 RNN layers of 400 size. Each layer uses an equivalent of 3 linear-layer-like matrix multiplications in GRU model. During inference, the input batch is a small number (usually 1) and thus, these neural networks require a ridiculous amount of bandwidth.

Now that you've understood that the choice of hardware comes down to the application that's going to be run, let's see who all are the major players in the AI chip manufacturing business. There's Intel with their Intel® Xeon™ Scalable Processor family, Intel® Movidius™ NCS, Intel® Stratix® FPGA, etc. Then there's NVIDIA with their TESLA GPUs. AMD isn't far behind with their Radeon Instinct lineup.

However, it's not just these three, there are many more companies producing AI hardware. There's Wave Computing, Cerebras Systems, Graphcore, DeePhi Tech, MediaTek, Samsung, etc., who have produced AI hardware. Even companies such as Amazon, Google, Microsoft and Apple have been working on specialised hardware for AI applications.

Building an AI model has two primary stages, there's the more intensive training phase and there's the not so intensive inference stage. For training, you can use CPUs such as the Intel® Xeon™ Scalable Processor family or GPUs such as those made by NVIDIA and AMD. When it comes to inferencing , there are hardware solutions such as Intel® Core™ processors with AVX2 instruction set, Intel® Atom™ Processors with the SSE instruction set or even the Arria A10 FPGA discrete hardware.

SOME DEEP LEARNING USE CASES

Since we're dealing with Intel systems in the mix, let's take a tour of the best hardware for different types of deep learning use cases.

Let's start with the popular x86 series that are typically multi-core (2 to 10 cores and can go up to 18 cores with the Intel® Core™ i9-7980XE and 32 cores in the AMD Ryzen Threadripper 2990WX. These are excellent configurations for deep learning projects that are centered on NLP